```
import random

board = ['O', ' ', ' ',
         ' ', 'O', 'X',
         'X', ' ', ' ']

def print_board():
    print(board[0] + ' | ' + board[1] + ' | ' + board[2])
    print('---------')
    print(board[3] + ' | ' + board[4] + ' | ' + board[5])
    print('---------')
    print(board[6] + ' | ' + board[7] + ' | ' + board[8])

def check_win(p):
    wins = [(0,1,2), (3,4,5), (6,7,8),
            (0,3,6), (1,4,7), (2,5,8),
            (0,4,8), (2,4,6)]

    return any(board[a] == board[b] == board[c] == p for a,b,c in wins)

while True:
    print_board()

    move = int(input("Choose a position (1-9): ")) - 1
    if board[move] != ' ':
        print("That spot is taken!")
        continue

    board[move] = 'X'

    if check_win('X'):
        print_board()
        print("You win!")
        break

    if ' ' not in board:
        print_board()
        print("Draw!")
        break


    empty = [i for i in range(9) if board[i] == ' ']
    ai_move = random.choice(empty)
    board[ai_move] = 'O'

    if check_win('O'):
        print_board()
        print("AI wins!")
        break
```

```
O |   |
---------
  | O | X
---------
X |   |
Choose a position (1-9): 9
O |   |
---------
  | O | X
---------
X | O | X
Choose a position (1-9): 3
O |   | X
---------
  | O | X
---------
X | O | X
You win!
```

```
def simple_reflex_agent(room_a, room_b, position):
    print("SIMPLE REFLEX AGENT")

    rooms = {
        "A": room_a,
        "B": room_b
    }

    while True:
        print("Vacuum is in Room", position)

        if rooms[position] == "D":
```

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            print("SUCK")
            rooms[position] = "C"

        elif position == "A":
            print("MOVE RIGHT")
            position = "B"

        elif position == "B":
            print("MOVE LEFT")
            position = "A"

        if rooms["A"] == "C" and rooms["B"] == "C":
            print("Both rooms are clean.")
            print("STOP")
            break

print("VACUUM CLEANER AGENT")

room_a = input("Enter status of Room A (D/C): ").upper()
room_b = input("Enter status of Room B (D/C): ").upper()

while room_a not in ["D", "C"] or room_b not in ["D", "C"]:
    print("Enter only D for Dirty or C for Clean.")
    room_a = input("Enter status of Room A (D/C): ").upper()
    room_b = input("Enter status of Room B (D/C): ").upper()

position = input("Enter starting position (A/B): ").upper()

while position not in ["A", "B"]:
    position = input("Enter starting position (A/B): ").upper()

print("Initial State:")
print("Room A:", room_a)
print("Room B:", room_b)
print("Vacuum Position:", position)

simple_reflex_agent(room_a, room_b, position)
```

```
VACUUM CLEANER AGENT
Enter status of Room A (D/C): D
Enter status of Room B (D/C): D
Enter starting position (A/B): A
Initial State:
Room A: D
Room B: D
Vacuum Position: A
SIMPLE REFLEX AGENT
Vacuum is in Room A
SUCK
Vacuum is in Room A
MOVE RIGHT
Vacuum is in Room B
SUCK
Both rooms are clean.
STOP
```